8.7 Chaos You Can Solve

WORKSHEET

The question

STEP I 2014, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Where have you seen the expression $4s(1-s)$ before, with $s = \sin^2\theta$? Which identity connects $\sin^2\theta$, $\cos^2\theta$ and $\sin 2\theta$?

STAGE 1 Guessing the pattern

Covers: Part (i), the conjecture

Ask yourself With $u_0 = \sin^2\theta$, what is $1 - u_0$? Write u_1 as a single square using the double-angle formula. Then do u_2.

STAGE 2 Proving it

Covers: Part (i), the proof

Ask yourself Prove your guess by induction on n. The inductive step is the same calculation as for u_1, with $2^n\theta$ in place of θ.

STAGE 3 Changing variables

Covers: Part (ii), the condition on p, q, r

Pause and try Substitute $v_n = \alpha u_n + \beta$ into the rule for v_{n+1} and rearrange for u_{n+1}. Match with $4u_n - 4u_n^2$: how many equations do you get, and how many unknowns? The leftover equation is the condition.

STAGE 4 The example

Covers: Part (ii), the example

Ask yourself Read off p, q, r and check the condition. Find α and β, and the starting value u_0. Which angle θ has $\sin^2\theta = u_0$?

What's really going on

Why the sequence is unpredictable

The formula $u_n = \sin^2(2^n\theta)$ looks like the end of the story: we can compute any term directly. But look at what it says. Since $\sin^2$ has period π, only the value of $2^n\theta$ modulo π matters. Write $\theta = \pi x$ with $0 \le x < 1$, and write x in binary:

$$x = 0.b_1b_2b_3\ldots \quad \text{(base 2)}.$$

Multiplying by 2 shifts the binary point one place to the right, and taking the result modulo 1 throws away the digit that crosses the point. So u_n depends only on the binary digits of x from b_{n+1} onwards. Each step of the logistic map reads one more digit of the starting value and discards one from the front.

This is *sensitive dependence on initial conditions*, the defining feature of chaos. Two starting values that agree to 20 binary places (about six decimal places) typically produce sequences that agree for about 20 steps and then have nothing to do with each other (Figure 8.7.1). To predict u_{100}, you need to know u_0 to 100 binary places. Any error in the starting angle, however small, doubles at every step. (An error in the population u itself does not grow by exactly the same factor each time, since the map stretches some parts of $[0,1]$ more than others, but on average it grows at the same rate.) The formula does not help: it simply makes the mechanism visible.

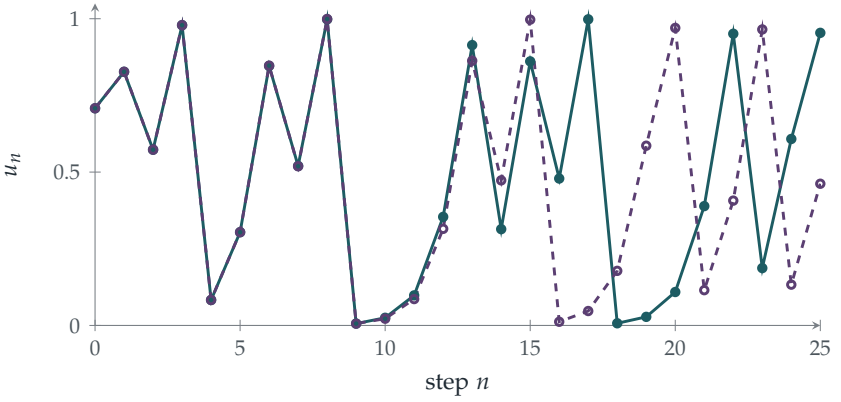


Figure 8.7.1 Two sequences of the logistic map with $k = 4$, starting from $\sin^2(1)$ (teal) and $\sin^2(1.00001)$ (plum, dashed). They are indistinguishable for about ten steps, visibly different by step 12, and unrelated after that: by step 12 the difference in angle, 10^{-5}, has been doubled twelve times, to about $2^{12} \times 10^{-5} \approx 0.04$.

Periodic points everywhere

The formula also finds every cycle of the map. The sequence returns to its starting value after N steps exactly when $\sin^2(2^N\theta) = \sin^2\theta$, that is, when $2^N\theta \equiv \pm\theta \pmod{\pi}$. So $\theta = m\pi/(2^N - 1)$ or $\theta = m\pi/(2^N + 1)$ gives a point of period dividing N. For example $\theta = \pi/5$ satisfies $4\theta \equiv -\theta$, so $\sin^2\frac{\pi}{5} \to \sin^2\frac{2\pi}{5} \to \sin^2\frac{4\pi}{5} = \sin^2\frac{\pi}{5}$ is a 2-cycle. There are periodic points of every period, and they are dense in $[0, 1]$: in every interval, however small, there is a starting value whose sequence repeats. Yet each one is unstable, because any error doubles. Dense periodic points that repel everything nearby are the second hallmark of chaos.

The road to chaos

For smaller k there is no formula, and the behaviour depends dramatically on k (Figure 8.7.2). For $1 < k < 3$ every sequence starting in $(0, 1)$ settles down to the fixed point $1 - 1/k$. At $k = 3$ the fixed point becomes unstable and a 2-cycle takes over (this is the subject of STEP II 2002 Q5; see the exercises). At $k = 1 + \sqrt{6} \approx 3.449$ the 2-cycle gives way to a 4-cycle, then an 8-cycle, with the doublings coming faster and faster, until at $k \approx 3.5699$ the cycles have infinite length and chaos begins. Inside the chaotic region are windows of order: the clearest, near $k = 3.83$, contains a 3-cycle, which is the subject of Section 8.8.

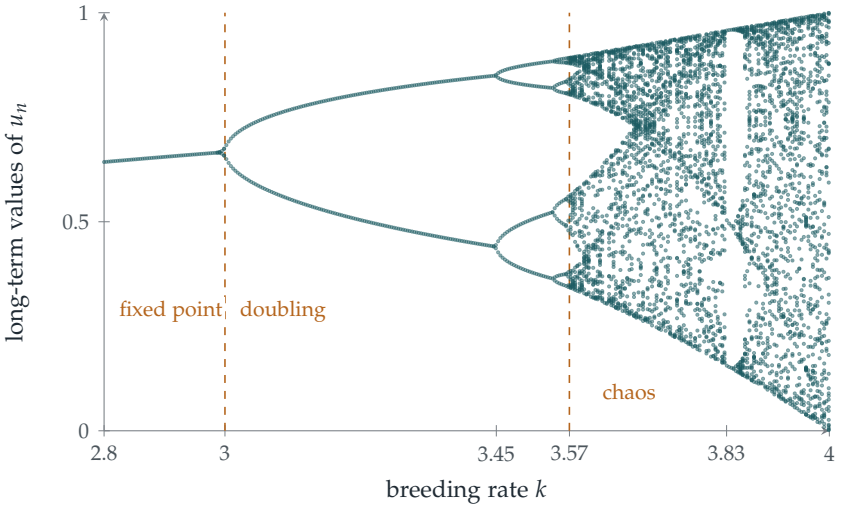


Figure 8.7.2 The bifurcation diagram of the logistic map. For each k, the map was iterated 300 times from $u_0 = 0.31416$ and the next 40 values plotted. A single fixed point splits into a 2-cycle at $k = 3$, then 4, 8, ..., until chaos sets in near $k = 3.57$. The pale vertical gap near $k = 3.83$ is the window containing a stable 3-cycle. At $k = 4$ the values fill all of $[0, 1]$.

Feigenbaum's constant

In 1975 Mitchell Feigenbaum, working at Los Alamos with a programmable pocket calculator, measured the values of k at which the period doubles: $k_1 = 3$, $k_2 \approx 3.4495$, $k_3 \approx 3.5441$, $k_4 \approx 3.5644$, ... He noticed that the gaps shrink by a constant factor,

$$\frac{k_n - k_{n-1}}{k_{n+1} - k_n} \to 4.6692\ldots,$$

and then, astonishingly, that the same number 4.6692... appears for $x \mapsto k\sin(\pi x)$ and for every other smooth map with a single hump that is rounded like a parabola at its top (a *quadratic maximum*). The constant is *universal* for that whole class; maps whose hump is flatter at the top, such as one shaped like $-x^4$, have universal constants of their own. It has since been measured in experiments on dripping taps, heated fluids and electronic circuits. Explaining why required ideas borrowed from the physics of phase transitions, and was only made rigorous by Oscar Lanford in 1982, with the help of a computer.

The same trick, three times

The substitution in part (ii) is an example of *conjugacy*: two maps are conjugate if a change of variable turns one into the other, and then they have exactly the same dynamics. The question shows that every quadratic map satisfying the condition is conjugate to the logistic map with $k = 4$, which in turn is angle doubling in disguise,

through $u = \sin^2\theta$. (Strictly, this last link is a *semiconjugacy*: θ and $\pi - \theta$ give the same u, so the change of variable is not one-to-one. The logistic map is genuinely conjugate to the *tent map*, $t \mapsto 2t$ for $t \le \frac{1}{2}$ and $t \mapsto 2 - 2t$ for $t \ge \frac{1}{2}$, through $u = \sin^2\frac{\pi t}{2}$, which is one-to-one on $[0, 1]$.) In Section 2.8 you met $x \mapsto x^2 - 2$, solved by $x = 2\cos\theta$ with the angle doubling at each step; it satisfies the condition too (try it). So these are all the same map in disguise. In Section 8.8, a third member of the family appears: a map built from straight lines, the *tent map*.

Try this next

1 **Cycles of the logistic map.** Try STEP II 2002, Question 5 (step.maths.org/questions/02-s 2-q5). Then show that a 2-cycle exists only when $k > 3$.

2 **Squaring minus two.** Show that $v_{n+1} = v_n^2 - 2$ satisfies the condition of part (ii) (write it as $-pv_n^2 + qv_n + r$ with $p = -1$). Find α and β, and check that your answer agrees with $v_n = 2\cos(2^n\phi)$.

3 **A 3-cycle.** Use the formula $u_n = \sin^2(2^n\theta)$ to find a 3-cycle of $u_{n+1} = 4u_n(1 - u_n)$. (*Hint:* try $\theta = \pi/7$, or $\theta = \pi/9$.) How many points have period exactly 3?

4 **Losing digits.** A computer stores numbers to about 53 binary places. If you iterate $u_{n+1} = 4u_n(1 - u_n)$ on a computer, roughly how many steps before the computed values bear no relation to the true sequence? Is the computed sequence then useless?

Further reading

- **Robert May, "Simple mathematical models with very complicated dynamics"**, *Nature* 261 (1976). A classic, still very readable.
- **James Gleick, *Chaos: Making a New Science*** (Penguin). The story of May, Feigenbaum and the discovery of chaos.
- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press), Chapter 10. One-dimensional maps, the logistic map and Feigenbaum's constant, at first-year undergraduate level.
- **Ian Stewart, *Does God Play Dice?*** (Penguin). A popular account of chaos, with a good chapter on the logistic map.

The full guided solution is in Section 8.7:
https://beyondstep.keithmcnulty.org/ch08/solvable-chaos.html